Kaden Diaz

Mid-Level AV Systems Engineer

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PROFILE

Mid-Level AV Systems Engineer focused on federal and enterprise AV/VTC, networked AV integration, signal-flow troubleshooting and technical mentoring. Founder and lead developer of Vector Learning, building infrastructure, cloud and AI engineering skills through personal labs and AI-assisted software projects. Active DoD Secret Security Clearance.

SELECTED ENGINEERING ACCOMPLISHMENT

Restored Crestron NVX video - Traced a video-routing failure to conflicting endpoint roles, corrected the encoder/decoder configuration and supported the technician through verification.

Completed troubleshooting case · owner-reported

PROFESSIONAL EXPERIENCE

Mid-Level AV Systems Engineer
Geospatial and Cloud Analytics (GCA)
June 2026 - Present

- Integrate and troubleshoot multi-vendor AV systems, networked endpoints, DSP, video distribution, cameras and conferencing technology.

- Use signal-flow, HDCP/EDID and connectivity diagnosis; support AV drawings, bills of materials, service coordination and technician mentoring.

Junior AV Technician / AV Technician Intern
Geospatial and Cloud Analytics (GCA)
May 2025 - June 2026 (includes a school-term break)

- Supported AV/VTC troubleshooting, equipment integration and networked AV endpoints.

- Built practical experience with control, conferencing, audio and video systems while progressing from intern to Junior AV Technician.

In-Class AV Assistant
Virginia Commonwealth University
September - December 2025

- Supported classroom AV, Zoom and Microsoft Teams hybrid sessions, instructors and Executive MBA sessions.

CORE COMPETENCIES

AV systems integration | AV-over-IP | Crestron | Extron | AMX | Dante | DSP | HDCP/EDID | Technical mentoring | Linux lab administration | KVM/QEMU | Python and TypeScript project development

Technical work & learning

Mid-Level AV Systems Engineer

SELECTED PROJECTS

OptiPlex Linux infrastructure - Personal lab · owner-reported
Ubuntu, SSH and KVM/QEMU lab. Prometheus, Grafana and Uptime Kuma installed, owner-reported pending artifacts; monitoring reliability remains unverified.

Vector Learning - Product in development
Founder and lead developer of a technical-learning platform. Public feature verification is ongoing.

AV Portfolio Suite - Locally tested prototype
AV planning calculations with 14 local calculation checks. Dashboard and commissioning data are simulated.

AV Integrity Lab - Locally tested prototype
PowerShell configuration-integrity lab and a separate Python companion with 19 local CLI checks. Original PowerShell execution remains unverified.

Cloud Task Suite - Locally tested prototype
Flask/SQLite learning application for duplicate review, protected records, atomic reservations and source-labeled retrieval. Improved local copy passed 34 application tests.

AV Operations Command Center - Platform prototype
Existing Base44 application with seven reviewed entity schemas for AV operations. Runtime acceptance and public deployment remain unverified.

Engineering portfolio - Static portfolio · locally tested
Next.js, React and TypeScript website with case studies, evidence filters and automated checks. Implemented with AI assistance; owner-led practice is recorded separately.

SELECTED CREDENTIALS & TRAINING

Owner-reported completed credentials and training. Professional certification, vendor training and coursework are identified separately.

- AVIXA CAVT (certification)
- Audinate Dante Level 3; Crestron DMC-E, Commercial and Residential Technician, Modern Workplace (vendor training)
- Extron ECS, NAV, ProDSP, Global Configurator Plus, GUI Designer, AV Associate (vendor training)
- AMX Technician and Designer; NETGEAR NCE, NCE-AV and AV Levels 1-3; Q-SYS Level 0 (vendor training)
- Sennheiser RF Basics; Cisco Network Technician Career Path; Microsoft Azure, AI, Security and Security Copilot coursework (training)

EDUCATION

J. Sargeant Reynolds Community College | Associate of Science - Social Sciences, May 2024

Virginia Commonwealth University | Computer Science coursework